

Interdisciplinary modeling and participatory simulation of forest management to foster adaptation to climate change

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ABSTRACT

The adaptive capacity of forests and foresters to overcome the adverse effects of climate change remains highly uncertain despite intense research efforts. While foresters are often invited “not to put all their eggs in one basket,” adaptation strategies to climate change mostly depend on silvicultural diversification. To explore how socioeconomic adaptive tools can complement these technical evolutions in forestry, we designed an interdisciplinary and participatory simulation of forest management combining a role-playing game, ecological models of forest evolution, and a severe climate change scenario. Participants from French natural parks and forest organizations responded positively to its multiple applications. Here, we investigate the technical and timber-focused framing of climate change by forest managers. We also analyze participants’ negotiations when attempting to change the simulation rules of forest management. Drawing on this experience, we highlight how establishing a payment system for ecosystem services can reduce financial imbalance driven by climate change.

1. Introduction

The reality of climate change is now certain, although the magnitude of its spatial and temporal impacts on the environment is not (IPCC, 2021). As one of the most important ecosystems on the planet, forests will face profound long-term changes such as modified species assemblages to an extent that is still unpredictable (García-Valdés et al., 2020; Lindner et al., 2010). In the short to medium term, the productivity of forests in southern and eastern Europe is expected to worsen because of the decreased health of forests caused by rarer cold extremes and more frequent heat waves (Bréda et al., 2006; Ciais et al., 2005; Lindner et al., 2010). The interaction of these hazards and phenological changes with introduced and native pathogens is another matter of concern, with a great potential for damage (Bakys et al., 2009; Futai, 2013). Hence, it is unclear whether forest species will have sufficient adaptive capacity to cope with climate change (Aitken et al., 2008; Corlett and Westcott, 2013). For instance, the ability of tree populations and species to overcome climate change is still being discussed (Alberto et al., 2013). As a result, forest managers are trying to develop complementary adaptation strategies (Keenan, 2015; Millar et al., 2007) to maintain the numerous ecosystem services (ES) provided by forests such as timber

production, leisure activities, carbon storage, and water filtering (Brockerhoff et al., 2017). A key feature of robust anthropogenic adaptation is accounting for the uncertainties of climate change: in a highly uncertain context, foresters can favor risk dilution by using different tree species, ages, and silvicultural systems (Kennedy and Koch, 2004; Naumann et al., 2011). Instead of optimizing the performance of a forest stand, this “bet-hedging” approach seeks to minimize ES loss (Spittlehouse and Stewart, 2004).

The issue of climate change is taken seriously in the field of forest sciences, and multifarious research projects aim to develop a series of readily available adaptation options (Bolte et al., 2009; Fouqueray and Frascaria-Lacoste, 2020; Keenan, 2015; Millar et al., 2007) or refine predictive models of forest growth under climate change (Härkönen et al., 2019). To assess the operability of these innovations, researchers can rely on an active network of forest landholders and managers eager to participate in experimental tests, as demonstrated by many studies conducted in France (CNPF, 2016), Europe (Kolström et al., 2011), and the United States (Nagel et al., 2017). However, there is a difference between studies that expand existing knowledge on the potential range of adaptation strategies and those that explore what is currently being undertaken in the field. The available literature on what foresters are

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actually doing to prepare for climate change is scarce. Based on individual interviews and questionnaires, it shows that field practitioners do not wait for research outcomes to change their management practices (Fouqueray et al., 2020; Kolström et al., 2011; Van Gameren, 2014). It also stresses the emphasis placed on timber production and forestry technical changes such as species replacement or the tree density of forest stands, which is a typical trend in forestry (Dobbertin and Nobis, 2010).

Two conclusions arise from this state of the art. First, the interactions between the different forest stakeholders in the adaptation process have been scarcely examined to date. When adapting, forest managers gear their silvicultural practices toward the provision of one or multiple target ES (Duncker et al., 2012), which can positively or negatively affect the management success of the neighboring stands: for instance, a preference for hunting in one place can threaten the regeneration of trees in another. A forester can also be inspired by management changes made by another forester, or they can develop ideas together that would not have emerged otherwise. Second, several articles stress the value of thinking outside the box and considering non-technical changes as complementary mechanisms (Hallegatte, 2009; Jacobs et al., 2015; Naumann et al., 2011). For instance, as foresters draw most of their financial resources from timber production, they will be at risk of climate change threatening the productivity of trees. To balance their sources of revenue, foresters can look toward insights from other disciplines as opposed to the forest sciences alone. For instance, they can look toward the social and economic sciences to adopt economic tools for timber sales by adding value to timber products through quality labels or turning toward insurance and supply contracts with climate-related clauses, among others. Foresters can also diversify their income by investing in ES other than timber production such as carbon storage contracts (Tronquet et al., 2017).

To the best of our knowledge, no forest management research has addressed the issue of adaptation to climate change with these two questions in mind. Consequently, the objective of this paper is to describe a new method addressing this knowledge gap and to present the results drawn from its application in nine different study cases. The first step was to develop a methodological tool able to do the following: (1) be prospective and integrate foresters' decisions at a regional level, because this is the scale at which most forest public policies are negotiated – at least in France, the country of the study cases (MAAF, 2017); (2) incorporate technical adaptations but artificially limit their effectiveness because of a strong climate change scenario encouraging foresters to develop complementary tools from the organizational or economic sciences; and (3) allow the emergence of spontaneous and/or collective designs for adaptations to climate change in forestry.

With these reasons in mind, we created *Foster Forest*, a participatory simulation combining a computerized agent-based model (ABM) and a role-playing game (RPG) as defined by Barreateau et al. (2001). Participatory simulations and RPGs tend to focus on existing forest issues such as non-temperate ecosystems or the interaction between forestry and other land uses (e.g., Etienne et al., 2008; Étienne, 2003; Fauvelle and Garcia, 2018), instead of adaptation to climate change. However, RPGs are pertinent for the study of adaptation to climate change (Reckien and Eisenack, 2013; van Pelt et al., 2015) and global changes associated with forests (Garcia, 2019). We decided to use the companion modeling (ComMod) approach (Barreateau et al., 2003; Étienne, 2010), a relevant methodology to develop a participatory simulation on forestry in temperate ecosystems. Indeed, it is used for research, training, or negotiation purposes and is particularly useful for encouraging participant understanding and engagement in the management of social-ecological ecosystems (SES) (Bousquet and Le Page, 2004; Voinov et al., 2018). ComMod relies on the involvement of stakeholders to define and develop a model of the SES of interest. Finally, since stakeholders involved in the design of the model do not necessarily participate in the simulation (Hassenforder et al., 2016), participants are invited to embody the different roles integrated in *Foster Forest*. They

make management decisions in a simulated hybrid environment that reproduces social, economic, and ecological processes (Becu et al., 2016). Some of these processes can be computerized (most often in an ABM): in the case of *Foster Forest*, this includes indicators of social satisfaction, prices for hunting rights, or forest growth. Lastly, ComMod complies with our methodological requirements, because it can model SES at a regional scale, allow spontaneous changes to emerge in the “rules of the game,” and explore scenarios of future forest management. To best apply the interdisciplinary ComMod approach, our research team was comprised of two ecologists, a geographer, and an economist.

In the following sections, we introduce the conceptual framework used to decide modeling choices. We then present the ABM and RPG behind *Foster Forest*'s participatory simulations. We conclude with the description and discussion of the results relating to the nine French study cases. France was chosen as our study area, because its forestry situation is particularly relevant to climate change adaptation: the country has the fourth largest area of forests in Europe (165,000 km²) (MAAF, 2017). Its legal regulations and public policies are also representative of countries adopting a multifunctional management of forests with the simultaneous provision of timber production, leisure activities, and biodiversity conservation (Légifrance, 2012). The timber industry and forest management provide numerous jobs, whether in private stands (75% of the surface) or in public forests (MAAF, 2017). Thus, adaptation is of great importance and is acknowledged in numerous public policies (MAAF, 2017; ONERC, 2015) and research projects (Fouqueray and Frascaria-Lacoste, 2020).

2. Collective construction of the conceptual model

2.1. A social-ecological framing of stakeholders' interdependencies

The analytical conceptual framework developed by Barnaud et al. (2018) for studying ES, social interdependencies, and collective action in SES inspired the design of *Foster Forest*. ES are at the core of this framework, which stresses the synergetic, antagonistic, or neutral relationships between ES providers and beneficiaries and their intermediaries (Fig. 1). The interdependencies can relate to different ES providers (e.g., private landholders and managers of public forests), ES providers and beneficiaries (e.g., forest managers whose stand regeneration suffers from overgrazing by boars and deer, hunters responsible for avoiding the overpopulation of game animals), and ES beneficiaries (e.g., motorcyclists, hikers). These interdependencies are reshaped by changes in management decisions that modify ES and by new trade-offs between the different interests of forest stakeholders. Intermediaries promoting collective actions such as natural regional parks (NRP) also contribute to the permanent rearrangement of these social relationships.

To apply this framework to *Foster Forest*, we decided that participants would have asymmetrical interdependencies (Becu, 2020). Whereas the information about the forest is freely accessible (land ownership, soil quality, etc.), each player has its own objectives and actions (Fig. 2).

2.2. Building the conceptual model

The conceptual model (Fig. 1) required the reduction of forest management complexity to its minimum features of interest. A major challenge was to balance this reduction with our objectives of studying adaptations to climate change, which potentially affects every single aspect of forestry. Another difficulty was to create an intermediate model of forestry: not too specific to a given region but also not too broad so that foresters could easily connect it to their reality.

Because of our specific requirements, we applied a shortened version of the ComMod approach to build the conceptual model. As we needed a generic tool applicable to any French temperate forest instead of a precise regional case, we did not involve stakeholders during the earliest stages of the modeling process. Instead, we completed a first version of the model using previous analyses of foresters' decision-making

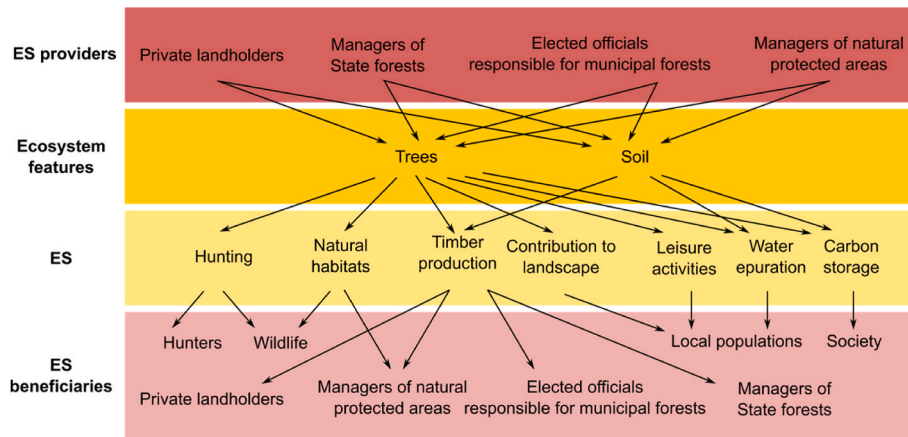


Fig. 1. A simplified version of the *Foster Forest* conceptual model. ES: ecosystem services. Forest ES that do not relate to adaptation issues such as erosion control were not included in the model.

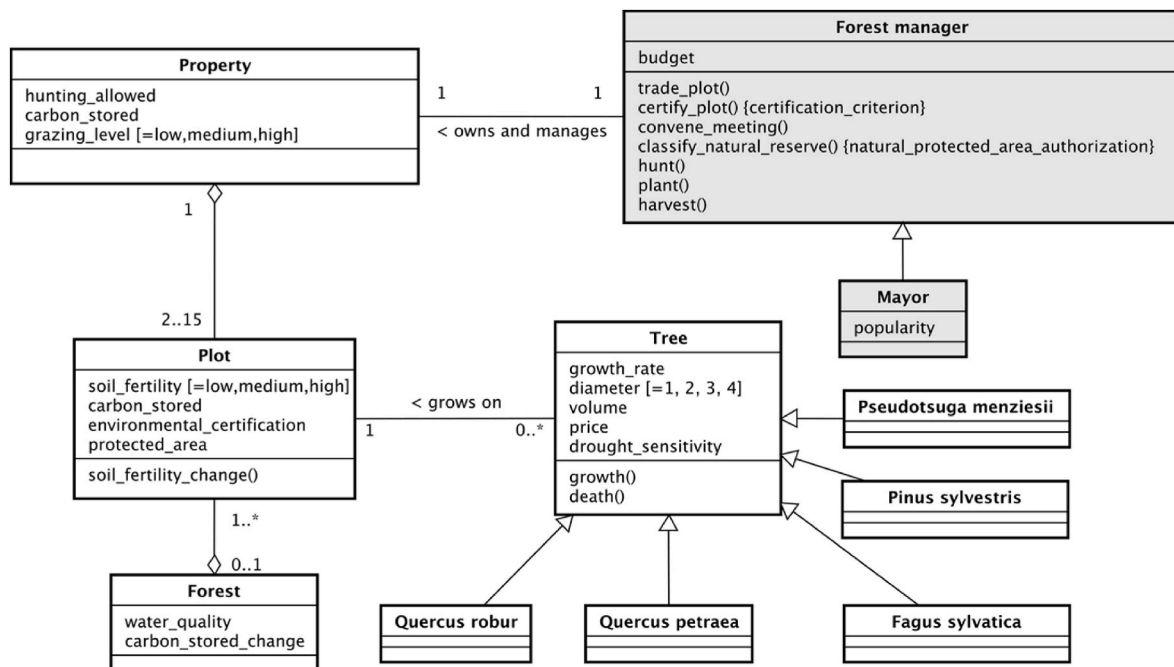


Fig. 2. UML overview of the *Foster Forest* agent-based model. Gray: non-autonomous agents controlled by *Foster Forest* participants.

(Fouqueray et al., 2020). Then, as a first step, we interviewed different forestry experts to gain complementary insights to develop an improved version. A second step led to the final version of the conceptual model, conceived during a test workshop hosted by the NRP of the “Vosges du Nord,” which brought together public and private foresters in September 2018.

The inclusion of forest stakeholders greatly improved the model, for instance, with regard to the exclusion of abrupt disturbance events. Fires, droughts, or windstorms are far from negligible, although cognitive biases increase the chances that they lead to adaptations compared to the incremental impacts of climate change (Morin et al., 2015; Weinstein, 1984). Foresters are used to thinking in the long term, so we tailored *Foster Forest* to progressively integrate the evolution of mean annual temperatures and precipitations among others. It was in this vein that one interviewee suggested discarding disturbance events to focus on “slow variables,” an opinion shared by the other participants of the test workshop. We followed his recommendation, which was consistent with feedback from other ComMod researchers that introducing a crisis would restrict the debates to short-term technical discussions among

participants. Participants of the game sessions generally agreed with these conceptual choices.

The second step was also valuable in order to legitimate certain biases such as making the participants the ultimate decision-makers of forest management. Intermediary bodies were not included as agents in the model; their influence was only considered in the evolutive processes of objects in the model. For instance, forest cooperatives can modify timber prices, although this effect is encompassed by random variations in timber prices around a fixed mean.

2.3. A model restricted to five providers of ecosystem services

The final conceptual model of *Foster Forest* includes five ES providers. Three are forest managers representative of publicly owned forests: an elected official of a municipality owning communal forests (“mayor”), a National Forests Office manager (“public forester” managing state forests), and a manager of a natural protected area (“protected area manager”). Two private forest landholders (“landowners”) complete the group. However, as it can be difficult to recruit landowners, they can be

replaced by a private forest advisor or a public advisor from the Regional Centers for Privately Owned Forests (regional public bodies seeking to increase wood mobilization among private landowners so as to indirectly stimulate the timber industry). During the participatory simulation, foresters play their own role.

In accordance with the conceptual model, each role is attributed certain objectives. The mayor must avoid the deterioration of water quality and promote forestry operations that positively contribute to the esthetic value of the forest landscape. Current regulations in French public forests aim to harvest 50% of tree production, which will certainly increase with the rising demand in forest biomass (Collectif, 2016; Scarlat et al., 2019). In line with this, the public forester must harvest 70% of the volume of the decadal tree production (a figure that the facilitator can adjust depending on the context of the workshop). The area manager must convince the other participants to subscribe to a program for the conservation of old-growth forest in order to reach a total of seven plots at the end of the simulation. She/he must also provide any relevant information likely to favor carbon storage. Like all the other players, the landowners must maintain a balanced budget and avoid the detrimental effects of overgrazing.

A facilitator hosts the participatory simulations by explaining the rules and playing the role of a climate expert or state official to respond to participants' questions about any climate, forest, or regulatory issue. She/he is accompanied by an observer in charge of noting information (discussions, behaviors, decisions, etc.) relevant to the research question.

The "Overview, Design Concepts, and Details" framework (Müller et al., 2013) summarizes *Foster Forest's* features in Appendix A to simplify its comparison with other ABM including human decision-making.

3. Agent-based model behind *Foster Forest*

Foster Forest is a hybrid composite simulation (Le Page et al., 2011), which combines features of both ABM and RPG. It is both hybrid and composite because of its combination of autonomous avatars of agents, humans controlling non-autonomous avatars (or their computerized representations), and semi-autonomous agents (humans with limited options of control). Here, we present the ABM on which *Foster Forest* relies.

3.1. Model classes and calibration sources

Foster Forest's ABM includes spatial entities (plots and properties), semi-autonomous social entities (foresters), and physical entities (trees) (Fig. 2). The simulation environment is stylized, which means that the characteristics of each of the classes are proportional to French temperate forests (Le Page and Perrotton, 2017). For instance, the initial distribution of pure or species-mixed forest stands is based on empirical data from the National Forest Inventory (IFN, 2017). Abstract and representative environments were discarded for two reasons. First, we needed a prospective tool to allow for comparisons between different regions in the same legislative and economic context, namely France. Abstract or representative environments could not lead to comparisons of regional social-ecological frameworks (Barreteau et al., 2001). Second, a stylized environment distances participants from the issues that they face on a daily basis and facilitates their reflective behaviors during the participatory simulation (Étienne, 2010).

Forest plots are the basic units of spatial entities. Plots have one parameter for soil fertility and another for carbon storage. Soil fertility, a combination of natural soil fertility and artificial soil compaction, is important for the calculation of tree growth. Updating soil fertility and aboveground carbon storage occurs at the plot scale. Depending on their tree species mixture, plots can be eligible for the old-growth conservation program. Plots are grouped by properties and distributed among foresters according to the mean proportions of the French ownership

structure, as taken from the National Forest Inventory (IFN, 2017). Properties are characterized by a level of grazing in order to account for the grazing pressure placed on tree regeneration by boars and deer (not represented in the ABM).

In the ABM, foresters are semi-autonomous social entities: participants control pre-determined actions of the "forest manager" class. Forest managers are characterized by forest ownership and a budget. They can trade forest plots or timber and decide on the intensity of hunting.

The only physical entities of the model are trees. Trees are located on plots and are categorized by species and diameter. The parameters of each species differ in terms of growth rate, wood price, and sensitivity to drought. We retained five different species, namely beech (*Fagus sylvatica*), pedunculate and sessile oaks (*Quercus robur* and *Quercus petraea*), pine (*Pinus sylvestris*), and Douglas fir (*Pseudotsuga menziesii*), so as to include a range of timber productivities, prices, and drought sensitivities similar to what foresters face in the field. For instance, beech was the most drought-sensitive species, while Douglas fir was the most profitable (Bréda et al., 2006; Direction des Ressources Forestières et al., 2017). Four categories of diameters were chosen to mimic the distinction of wood products used in the timber industry and refine the ecological parameters depending on the development of trees (e.g., young and old trees differ in terms of their survival, growth, and reproduction rates).

Appendices B and C present the parameter choices and data sources for the initial situation.

3.2. Computerized processes and players' actions

In *Foster Forest*, computerized processes calculate the successive stages of the spatial, social, and physical entities. Most function in response to the decisions taken by participants, although some form part of a fixed climatic or economic scenario. This could seem to go against the importance of having uncertain climate change scenarios in *Foster Forest*. However, the climate information given to the participants is very vague and only runs for the ongoing round. As first-time players, the evolution of the climate remains very uncertain from their perspective.

3.2.1. Simulation of forest growth

The survival, reproduction, and growth of trees are simulated at the stand (plot) level and not at the tree level in order to avoid cumulated model errors from the tree to the stand level (Porté and Bartelink, 2002). The inclusion of ecological processes such as asymmetric competition, the increased impact of grazing on seedlings, and so on (Fig. 3) was an important requirement in the selection process of a forest growth model for *Foster Forest*. The level of accuracy and the limitations of the growth model were communicated to participants during an introductory briefing.

The forest growth model used in *Foster Forest's* ABM is inspired by Kohyama and Takada (2012) as well as Mathias et al. (2015). The growth model, presented in Appendix B, simulates the dynamics of 1 ha of uneven-aged forest stands containing five different tree species of four diameters. Its execution by CORMAS was very rapid, which avoided interruptions during the participatory simulations.

We fixed the computational timestep at 10 years. After discussions with private and public foresters, this was considered to be a plausible balance between the 10-to-20-year duration used in private forests' management plans (mandatory for forest estates larger than 25 ha), the updating of climate parameters (further detailed in section 3 of Appendix B) and economic prices (see Appendix C), and the turnover rate of silvicultural operations in even-aged stands.

3.2.2. Computerizing participants' decisions and their effects on the agent-based model

Attendees can choose between many actions specific to forest

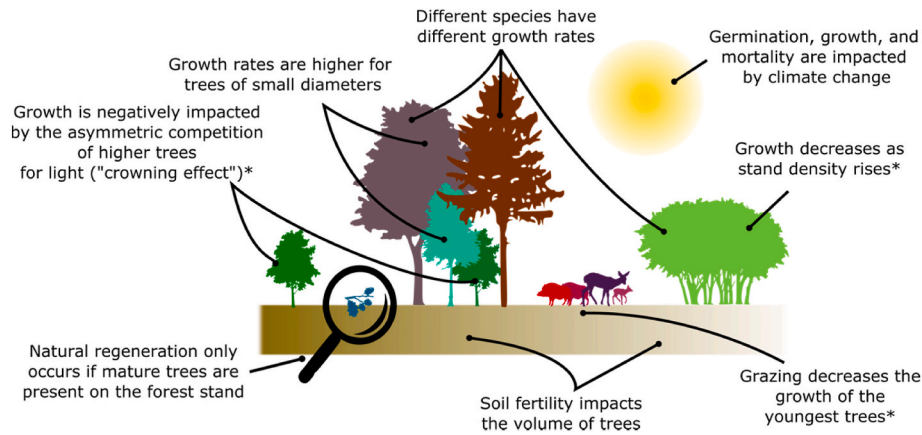


Fig. 3. Parameters influencing the growth of trees in *Foster Forest*.

*These parameters are species-specific. For instance, the crowning effect is higher for oaks than for beeches (Appendix B).

management such as tree harvesting or hunting. During a participatory simulation, players have two information sources at their disposal to reach their objectives. First, a personal booklet lists the different silvicultural actions (Fig. 2) and their consequences on social, ecological, and economic indicators (inhabitants' satisfaction; soil fertility, water quality, grazing rates; personal budget). Second, two maps of the simulation environment are projected (Fig. 4). One is projected onto a vertical screen; it shows the 50 plots of the simulation, with a background map of the property boundaries and a forest inventory of each plot's forest (number of trees per hectare, by species and diameter). The second map is displayed horizontally on a table close to the players using an ultra-short throw projector. It is made available at the request of participants to show them any information of interest. For instance, the protected area manager can ask for the location of plots eligible for the conservation program.

As soon as they decide on their forest operations, the participants go to the facilitator who implements their decisions on the computer.

Participants first specify their preferred contract for hunting leases, geared toward either hunting (a lower hunting intensity entails a higher presence of game with a higher grazing rate but also a higher lease payment) or timber production (lower lease payment but increased hunting and therefore lower grazing intensity affecting planted and natural seedlings) (Appendix B).

Participants then give instructions for the forest operations to take place on every plot of the estate that they possess or manage: voluntary or paid participation in the fake "PFSC" certification system – an *ad hoc* mixture of the Program for the Endorsement of Forest Certification (PEFC) and the Forest Stewardship Council (FSC) –, number of trees harvested (detailed by species and diameters), slash removal, and plantation if needed. The foresters' budgets change directly in line with the implemented silvicultural actions in order to reflect the costs (workforce wages, nursery seedlings, skidding, taxes, etc.) and benefits (timber revenues) of the forestry operations. Ecological indicators also evolve in relation to the participants' silvicultural choices: soil fertility decreases in the case of slash removal, clearcutting of an entire plot, or softwood planting, although it increases when nothing happens (Appendix C). Slash removal also lessens the belowground carbon stock (Appendix C). Water quality, measured at the whole forest scale, depends on the mean soil fertility, the proportion of coniferous species, and the number of clearcut plots (Appendix C).

In a last step, participants state their potential interest in selling or buying forest plots or in engaging in the old-growth conservation program. Land transactions occur at the end of the round, once all participants have computerized their decisions. The conservation program, which is operated by the protected area manager, targets the protection of old-growth trees. Conservation contracts last for 30 years, and during

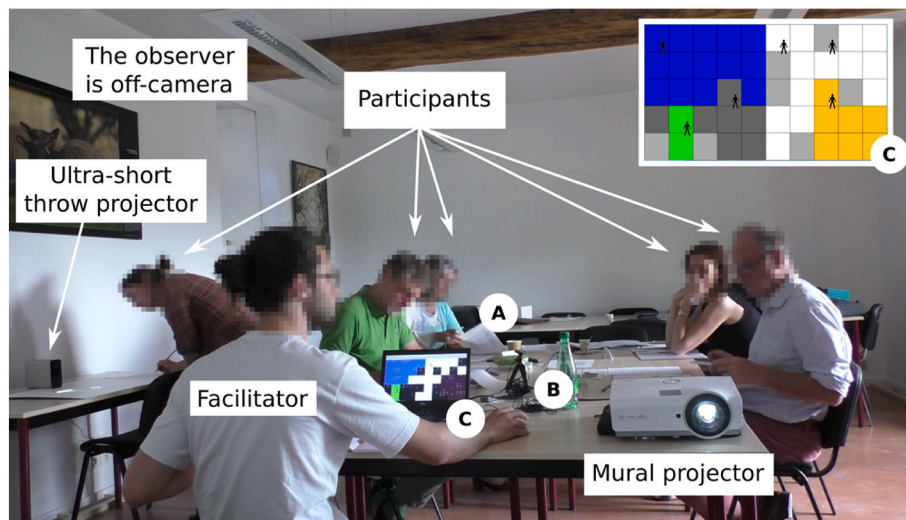


Fig. 4. Set-up of a participatory simulation of *Foster Forest*. A: The participant reads the explanatory booklet. B: Sound recording system, supplemented by a video recording system from which the photograph was taken. C: Background map of the simulation with colored properties. This main map is duplicated by a vertical projection onto the wall. Appendix D further details the visual interface.

this time, contractors engage not to cut the oldest trees. In compensation, they benefit from a payment proportional to the financial shortfall.

Appendix D displays two visual interfaces. The first is used to implement players' decisions during a round. The second presents the evolution of social, ecological, and economic indicators between two rounds.

3.3. Testing the calibration and scenarios

It was extremely complex to configure the calibration, scenarios, and rules to obtain a participatory simulation that was plausible, playable, and open to the spontaneous and novel ideas of participants. Therefore, the parameters and scenarios used in *Foster Forest* were tested during two preliminary sessions held with students and researchers prior to the nine study cases. The two test sessions brought about three major changes.

First, they highlighted the importance of an exhaustive introductory brief to justify the simplifications made to the conceptual model. For instance, the facilitator subsequently stated that land prices would not evolve and that the mayor's role would not be affected by municipal elections that would "occur" during a simulation.

Second, the test sessions emphasized the usefulness of pre-coded functions in the ABM, which were not communicated to participants, so as to facilitate a *free-play* mode permitting experimentation (Becu, 2020). As one of the objectives of *Foster Forest* is to encourage foresters to propose organizational or economic changes in their forest management, it helped the facilitator to have pre-coded, ready-to-use processes at his disposal to implement the players' ideas. However, these were only revealed if the foresters explicitly asked for them so as not to impede any spontaneous innovations that they could generate. Such functions included the calculation of carbon flows in every participant's property or the possibility for protected area managers to create a subsidy system to reward sustainable forestry practices.

Third, feedback also contributed to improving the graphical display of the available information sources as well as the spatial and temporal configurations of the simulations.

4. Foster Forest, a participatory simulation with a role-playing game

4.1. Temporal organization of the participatory simulations

The participatory simulation follows five steps: (1) a pre-simulation questionnaire; (2) a briefing; (3) different rounds of the simulation; (4) a post-simulation questionnaire; and (5) a collective debriefing.

The pre-simulation questionnaire aims to gauge the participant's sensitivity to climate change (e.g., if they had already experienced a climate catastrophe) and the multiple issues affecting forest management (e.g., the most important forest ES issues). The questionnaire also includes a "participation contract," which details the context of the study and its funding (public research, with no declared conflict of interest). By signing the contract, the players acknowledge that their participation is not binding. They give their consent for the authors to use their anonymized image and voice for the purpose of the research project.

During the introductory brief, the facilitator presents the different roles and their objectives, the information sources (booklets, mural and horizontal projections, and economic and climatic information posted on a wall), the list of possible actions, and the implementation process in the computer.

The simulation only begins after completing these two initial steps. It consists of three to five similar steps, each corresponding to a 10-year period (step 1: 2020–2029, step 2: 2030–2039, etc.). Each step starts with announcements concerning decadal projections for climate change as well as timber and carbon prices. The players can then freely talk with each other and approach the facilitator at any time to implement their forest management decisions, with no predefined order for taking turns.

A ringtone announces the end of the action period and the launch of the computerized updating of the spatial, social, and physical entities. At this point, the facilitator spends 10 min providing a collective overview of the forest, property, and stand parameters (see Fig. 2), and communicates the climate record of the last decade. The end of a step is also the appropriate time for participants to ask for the organization of a meeting on any issue of importance to them – for instance, a meeting was once requested by a mayor to elaborate a collective strategy to reinforce the social acceptability of tree removal. Overall, each step lasts 20–40 min, except for the first one (1 h) to account for the learning phase.

The post-simulation questionnaire aims to compare the strategies adopted during the game with the real-world choices operated by the foresters. It also asks participants what they learnt from the workshop and includes time for the suggestion of improvements.

Finally, at least 1 h is dedicated to collective debriefing. Debriefing is crucial, because it allows participants to provide feedback on the conceptual model and its implementation, which enables its validation and further improvements (Guyot and Honiden, 2006). The first part of the debriefing is inspired by the "most significant change" method (Perez et al., 2010): participants are successively encouraged to express one or two changes considered to be of particular importance to the simulation. This is experienced as a sort of emotional relief for the speaker, while it reminds other players what happened and provides insights into the speaker's priorities for subsequent analysis. This is also useful for the facilitator during the second part of the debriefing, which involves a roundtable discussion focusing on three topics. First, we ask participants how they tried to reach their objectives and what obstacles they encountered. Second, we return to the thought-provoking elements of the discussions or the spontaneous collective behavior relating to the issue of climate change adaptation. Third, we steer the discussion toward participants' reactions to the strong climate scenario, which often ended in a general discussion about the limitations and merits of participatory simulations.

4.2. Spatial organization of the participatory simulations

Seat distribution is unchanged throughout the simulations in order to facilitate the subsequent video comparison of strategies developed by different participants with the same role. The spatial configuration of seats also mimics real-world patterns of interactions between participants. For instance, it is legally embedded that the National Forests Office must provide technical assistance to elected officials of municipalities that own communal forests; hence, the public forester and the mayor are always placed next to each other.

When seated, all participants can see the vertical projection of the main map. However, they must move to see the horizontal screen, most often placed on a smaller support isolated from the central table, while the economic prices are pinned on a wall. This enables us to refine our observations regarding the information searched by the participants (who they are, when they need the information, etc.). Participants can also move throughout the simulation space to communicate with each other or with the facilitator. The computer on which the facilitator inputs the players' decisions is located on one side of the room. The observer does not interact with the participants.

To allow for the later analysis of the workshops, every simulation is recorded using a video camera and two voice recorders placed on the main table.

5. Results from nine study cases

5.1. Study cases analysis

5.1.1. Presentation of the study cases

Nine workshops, conducted following the methods described in section 3, occurred between May and July 2019. In chronological order, the simulations took place in the NRP of the "Vosges du Nord," in Rouen

(Normandy, two sessions), in the NRP of the “Pyrénées Ariégeoises” (two sessions), in the NRP of the “Perche,” in the NRP of the “Boucles de Seine Normande,” in Louviers (Normandy), and in the NRP of the “Volcans d’Auvergne.” The same facilitator-observer duo was present for the nine study cases.

Natural regional parks and the regional community forest federation of Normandy were essential partners to recruit players and host the simulations. Workshops were programmed to last for 4 h in order to facilitate the recruitment of participants who often have busy schedules. Organizers’ knowledge of local foresters and their involvement in the organization of the simulations greatly increased the legitimacy of the research team in the eyes of participants. Their interest in the tool was motivated by its potential use as a support for collective decision-making or awareness-raising about climate change (Flood et al., 2018). This was yet another demonstration of the joint use of ComMod approaches for research, training, or negotiation purposes (Barreteau et al., 2003).

5.1.2. Methodological approach

The analysis was based on the observer’s notes, the questionnaires, and the audio and video content collected during the workshops. Any moments of interest (e.g., discussions, negotiations) identified in the simulations were transcribed, as were all the debriefing sessions. Audio recordings and written transcriptions were analyzed using Sonal, an encoder software that conducts audio-textual synchronization (Nicolas, 2013). Sonal allows for thematic and statistic discourse studies such as topic occurrences and co-occurrences (e.g., “climate change” and/or “timber production”). For our analysis, we coded the informative segments of each participatory simulation with thematic tags. These tags related to the components of the conceptual framework (ES providers and beneficiaries, ecosystem features, ES, as detailed in Fig. 1), climate change features, and *Foster Forest* as a participatory simulation (Table 1).

Table 1
Analytical focus of the participatory simulations. The list is not exhaustive.

Analytical focus	Categories	Examples
Conceptual model	Providers of ecosystem services	National Forests Office Municipality owning communal forests Managers of natural protected areas Forest landholders
	Beneficiaries of ecosystem services	Same as the providers Local populations
	Ecosystem services	Timber production Water filtering Hunting Hiking, bike riding, etc. Carbon storage
	Interaction between ES stakeholders	Intermediary bodies (natural regional parks, etc.) Contractualization, public policies, collective action
Adaptive change in forest management	Forest management approach, <i>sensu</i> Duncker et al. (2012)	Free evolution Low to intense interventions Short term/long term
	Adaptation implementation	Effectiveness of the adaptation Cause of adaptation (incremental changes, climate projections, etc.)
Participatory simulation	Structuration and biases of the model	Climate forcing Hidden functions encoded in the ABM Individual/collective gaming
	Suggested improvements	Ergonomics Model reductionism Calibration

5.2. Lessons learnt from the study cases

5.2.1. Climate change adaptation: technical, timber-focused, and of secondary importance

Despite the strong emphasis placed on the impacts of climate change through the announcements of decadal climate projections, participants’ two primary matters of concern were the balance between forest regeneration and game animals, and the local population’s perception of forest operations. Public and private foresters regretted the many complaints made by forest neighbors about the modifications to forest scenery. Consistent throughout the study cases, this feeling of social pressure was most prominent in the “Volcans d’Auvergne” NRP, a very touristic region listed as World Heritage (UNESCO, 2019). The issue was also of importance in the “Boucles de Seine Normande” session, where players spontaneously called a meeting to create an awareness program, which the facilitator then directly encoded into the program by decreasing inhabitants’ satisfaction threshold.

In comparison to these two issues, participants’ verbal reactions to climate change were lesser than expected, and overall, they focused on the low rate of natural regeneration. We assume that this originates from the climate scenario, centered on incremental changes (less prone to trigger reactions) and not on strong climate events – by contrast, participants spoke at length about the disease outbreaks and windthrows that they had experienced, but this was not reflected in the simulations. Nevertheless, in all the participatory simulations, adaptive changes were notable in participants’ forest management decisions. In coherence with the findings outlined in the introduction, these changes focused on species replacement and stand density in order to maintain timber production. The essence of this belief in the potency of technical developments is epitomized as the search for “the right tree at the right place, even in future conditions,” as repeated in seven different workshops. Its best illustration is the quest for a turnkey solution: private foresters turn to public foresters to ask for “ready-to-plant” species; public foresters turn to foreign experiences for feedback or to the facilitator to “call the national scientific research center” in order to find drought-resistant tree species; and the facilitator aims to foster non-technical adaptations to climate change.

The search for adapted species was frequently shared by both public and private foresters, but some foresters considered that achieving such outcomes would take decades and that monospecific stands would not be resilient to parasite outbreaks. Hence, they favored risk-dilution through species mixtures. Even though this adaptation strategy was tailored to the stand scale, it was tree-centered, as virtually no forester spoke of forest fauna, flora, or fungi – the only mention of soil fungi was made by a naturalist not trained in forestry.

During the debriefing sessions, participants’ emphasis on their way of playing “like in the real world” and the similarity between our findings and insights from field studies (Fouqueray et al., 2020; Kolström et al., 2011; Van Gameren, 2014) stress the relevance of the subsequent analyses for real, non-game situations.

5.2.2. Socio-economic changes in the provision of ecosystem services

While the abovementioned adaptations were common to all workshops, innovative approaches to the conceptual framework of *Foster Forest* emanated from the study areas of Rouen, the “Pyrénées Ariégeoises,” and the “Perche.” The changes stemmed from the observation that foresters’ revenue sources mainly derive from two ES, namely timber production and hunting, which are both threatened by climate change. Indeed, the climatic scenarios introduced considerable uncertainty in terms of the ability of these ES to balance foresters’ budgets. In response, participants developed a system of payment for ecosystem services (PES) to attain financial equilibrium through other ES (in all three study areas, this related to carbon storage). As explained by the “Perche” private forest advisor, “I relied on [carbon] contracts to diversify my income and try not to be too dependent on timber prices.”

Interestingly, PES were chiefly instigated by participants who were

protected area managers but not professional foresters in real life. Based on the debriefings and their questionnaire responses, we may argue that they were inspired by PES pilot projects conducted in French regions, which they had learnt about from forestry newsletters and professional meetings. Collective negotiations addressed three of the four potential flaws of carbon contracts (i.e., heterogeneity of stand conditions, uncertainty of storage due to stochastic weather conditions, additionality of the project, but not the permanence of carbon storage) (Gren and Aklilu, 2016). Heterogeneity and uncertainty were addressed by designing financial incentives with an obligation of means but not of outcomes – most often, remuneration for species mixtures and irregular shelterwood forest management. As a consequence, permanence was not considered in the contracts, because the financial compensation was linked to the duration of the change in forest management and not to the duration of carbon storage. Additionality was evaluated on a case-by-case basis through discussions between protected area managers who oversaw payments for carbon storage and candidates for the contracts.

In the three workshops, the design of the payment for carbon storage allowed us to clearly distinguish between ES providers. Some private foresters adopted an economic approach by switching from being an ES provider to an ES beneficiary: “The day that I’m paid 50€ per hectare annually, I will plant beeches, hazel trees, and even banana trees if you want me to!” Alternatively, the rest of the private foresters relied on carbon contracts as transitional funding to adapt their forest stands to climate change, for instance, by planting drought-resistant species. Mayors and public managers regretted that carbon contracts primarily targeted private landholders, a choice that the protected area managers justified by the importance of private forests (put together, fragmented private plots represent 75% of forest surfaces in France and also in the model). Mayors and public managers stated their preference for large-scale collective planning of forest management. For spatial scales, they suggested the common management of hunting or water quality. For temporal scales, protected area managers relied on carbon contracts as a stepping stone to make forest stands eligible for a conservation program to establish a network of old-growth trees. During the debriefings, two protected area managers stressed the long-term effect of carbon or conservation contracts. In their eyes, even an unsuccessful contract negotiation provides an opportunity, because it raises owners’ awareness about conservation issues.

Spontaneous rule changes aimed at adapting to the scenario’s uncertainties mostly related to PSE. Non-economic narratives exist to provide complementary adaptive tools (Röling and Maarleveld, 1999), but they did not lead to any evolution in the roles of ES providers and beneficiaries, or their interactions. A first lever of change that was discussed was raising awareness about climate change, as acknowledged by private foresters: “The ideal thing is to apply [the clauses of the contract] but without a contract.” Legal tools were another lever of change. The recent possibility for any private landowner to engage in conservation easements (Légifrance, 2016) was briefly discussed but only on one occasion.

5.2.3. Participants’ feedback on *Foster Forest*

Feedback was gathered from the debriefing sessions and the post-simulation questionnaires. Overall, participants appreciated the workshops and validated the conceptual model. Despite lengthy debates on soil, growth, and the calibration of climate parameters, they found the reductionist level to be appropriate: “It’s not just a game with fancy hypotheses, it’s quite realistic and corresponds to reality.” They enjoyed the hidden functions and the facilitator’s ability to dig into the source code when required, for instance, to incorporate the abovementioned awareness program (section 5.2.1.).

They also proved that strategic choices are not predetermined. Despite using the same tool (conservation program), protected area managers adopted different strategies in the study cases. Some of them initially negotiated conservation contracts with private owners, because

they considered that they would be more responsive to the economic argument, while others engaged in discussions with public foresters, as they managed larger forests. We observed a third strategy, with managers applying the conservation program to themselves in order to receive financial income. The revenues were subsequently reinjected into the acquisition of forest plots for new conservation programs.

Most of the improvement requests related to the implementation of the ABM and RPG. Some of the biases voluntarily introduced into the simulations were relevant (e.g., spatial configuration of participants’ seats). However, participants found that other biases interfered with the simulations such as the large amount of technical information in the introductory booklets, which aimed at reproducing the asymmetrical understanding of forest functioning, since professional foresters, unlike mayors and some private owners, are used to dealing with technical information. However, one mayor summed up the situation: “We’re not experts!” By contrast, this technical bias highlighted the importance of sensory perception for forest management. A forest manager stated that graphical projections were “very mathematical, with colors and figures. I was very perturbed at the beginning, as I don’t see forests in this way.” Despite some complaints, this validated our use of a stylized environment to spatially represent forests as an assemblage of multifunctional space, a way to stimulate innovative and integrative solutions (Barnaud et al., 2013). Participants partially explained their lack of reaction to climatic announcements by the absence of pictures or drawings exemplifying abrupt climate change, a deliberate choice made in order to focus on slow-paced climate change. Drawing on participants’ suggestions for improvement, we introduced a forest fire event in a second educational version of the game. To date, however, we did not observe significant changes to the adaptive paths taken between the two versions of *Foster Forest*.

The sessions demonstrated the importance of collective discussions to open up perspectives on the incorporation of climate change in forest management. “We need everybody to face up to this challenge,” said a Pyrenean landowner, a conclusion that is consistent with similar findings on environmental issues (e.g., Redpath et al., 2018). Most importantly, the simulations emphasized the prominent role accorded to the intermediaries in designing and promoting collective adaptation to climate change. In *Foster Forest*, the rearrangement of the relationships between ES providers and beneficiaries was always guided by protected area managers and mayors. From a reflective point of view, the organization of collective workshops in the “real world” has also been endorsed by federations of community forests and natural regional parks, as detailed by Bertrand and Fouqueray (2017).

6. Conclusion

Through an interdisciplinary participatory simulation, this paper aims to address the shortcomings in the existing literature regarding the design of diversified, quickly deployable adaptations to climate change in forestry. In *Foster Forest*, forest stakeholders interact with each other in participatory simulations combining a role-playing game and an agent-based model in order to achieve multifunctional objectives of forest management. To overcome the adverse effects of not knowing the climatic and economic scenarios, they can rely on a fixed set of technical tools with limited effectiveness and/or develop collective forms of socio-economic action based on their discussions.

A challenging achievement in the design process was to find an appropriate balance between the reductionism of the conceptual model and its implementation in the ABM and RPG so as to allow players to act on any ES that they would find of interest. As shown by the spontaneous development of game changes in four workshops, this balance was achieved. The ability to experimentally test changes in forest management, with no real risk of financial or temporal loss, is one of the most appreciated features of *Foster Forest* and ComMod approaches in general (Étienne, 2010).

Our findings pave the way for further investigations, especially in

terms of the role of participation-based learning. Because of the uncertain nature of climate change, learning by trial and error is not feasible. Participatory simulations can compensate the impossibility of “learning by doing” by means of learning by simulating (Barreteau et al., 2001). We now aim to monitor whether *Foster Forest*'s participants developed other kinds of learning after contributing to such a serious game (i.e., normative, relational, or cognitive learning, as reported in den Haan and van der Voort, 2018). It is worth investigating if simulation-based learning will have tangible effects on participants' forest management, especially since the organizers from the federations of community forests and some natural regional parks stated that they have gained experience in the use of participatory simulations. Their willingness to collaboratively enlarge their toolbox for adaptation has come true, as we developed a simpler and more comprehensive version of *Foster Forest* to foster collective responses of forest management under a changing climate, now available at www.fosterforest.fr.

Software availability

Software name: FosterForest_III. First available: 2020. Program size: 2.3 MB. Program language: SmallTalk. Developer: Timothée Fouqueray (see contact information in the authors' section). Hardware: PC platforms supporting VisualWorks (VW7.6). Software: Cormas platform, version 2018.5.30 (download from <http://cormas.cirad.fr/en/outil/download>). License: Foster Forest is licensed with the French Agency for Program Protection. Freely available at www.fosterforest.fr. ODD+D description: see Appendix A.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envsoft.2022.105338>.

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